

Evolutionary adaptation in autonomous agent systems — a paradigm for the emerging enterprise

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The enterprise of the future needs to be flexible, adaptive, and to respond rapidly to market demands and opportunities. This, in turn, demands a flexible, adaptive infrastructure. A plausible way to achieve such an infrastructure is to base its organisation on autonomous agents, where each agent responds locally to the intra- and extra-organisational forces it experiences. Enterprise adaptation then arises as an emergent effect. The theories of Holland, and related work on complex adaptive systems, have a direct bearing on the emergence of system adaptation from the interaction of locally adaptive entities. This paper ties Holland's theory of artificial and natural adaptive systems to autonomous agent systems. To illustrate the use of these theories in this context, the paper presents a system of evolving autonomous agents, interacting in a simulated producer/consumer economic world. The paper presents preliminary results with the producer/consumer, evolving agents system, and discusses the implications of these results for evolving, adaptive enterprise infrastructure systems.

1. Introduction

The work described in this paper is an initial exploration of the potential synergy obtained from combining evolutionary computing (EC) techniques with those of market-based multi-agent systems. One of the authors [1] has developed an approach to EC in which the genetic mechanisms are embodied in the agents rather than a god-like central mechanism. This is more faithful to biological systems from which EC takes its inspirations. It also maps very well on to message-passing multi-agent systems such as those under development in a number of areas [2, 3]. In the EC framework, the agents exchange 'plumage' objects (which are effectively messages advertising capabilities), and genetic material objects (which are effectively messages describing sets of agent features, by means of which the capabilities can be achieved). In the female role, an agent uses genetic operators to combine her features with those described in the message, to come up with a specification for her offspring. In cases where the mother dies and is replaced by a single child, the maternal line can be thought of as a single immortal evolving agent.

The Dynamo¹ team in BT's Intelligent Business Systems Research Group is interested in multi-agent systems as a means of reducing the complexity of the software infrastructure supporting BT's business process, yielding decentralised systems that are robust, adaptive and scalable. In particular, we are interested in systems in which the agents interact by buying and selling services, and hence form a market (see Kearney and Merlat [4]). Market economies are complex dynamic systems with many

parallels to biological ecosystems, and there is considerable scope for inter-disciplinary cross-fertilisation.

This paper addresses this synergy primarily from an EC research perspective. From this point of view, a market-based agent system is a valuable laboratory, offering richness of behaviour from a relatively simple system. However, it is important to remember a complementary perspective — that of incorporating EC-based mechanisms into agent-based decentralised business applications. The paper can be divided into two main parts — firstly, the motivating ideas are discussed in more detail; then the experimental system is described and initial results presented. A few pointers to future research avenues are then presented.

2. The requirement for adaptation and emergence

In common with other large organisations, BT's business, products, and network infrastructure abound with requirements for the management of distributed systems. Currently, one of two extreme approaches is adopted.

- Centralised control

This has the advantage that coherent changes can be made to the whole system easily. One disadvantage is that such an approach is rigid and demands uniformity from the distributed elements. Another is lack of

¹ Dynamo stands for the Dynamics of Adaptive Market-based Organisations.

scalability — as the number of distributed elements is increased, a point is inevitably reached where the central bottle-neck causes problems.

- Independent control with communication

This has the advantage that each distributed element can be controlled separately as required to serve local needs. The disadvantage is that without co-ordination, there is no guarantee that the independent decisions will not conflict.

Examples of both are found in business process support software. Workflow management systems, from vendors such as Oracle, IBM, Plexus, etc, provide a means by which standard business processes may be defined centrally. Execution of these processes is then co-ordinated by routing intermediate results to people or software that can perform the next step as specified in the process definition. At the other end of the scale, there are individuals performing tasks independently, possibly using different application software, co-ordinating their activities by informal means such as e-mail, groupware, telephone, etc.

Clearly, an approach is required that achieves the benefits of both the above without the disadvantages, by combining local autonomy with coherence of the overall distributed system. This is what we mean by a decentralised approach, though Malone [5] would term it connected decentralised. This is why there is considerable interest in agent-based software. The key qualities that agent-based components and systems exhibit are: autonomy, reactivity, proactivity, and social behaviour [6, 7]. To achieve goals or maximise some measure of value (autonomy), agents act on their environment (proactivity). Conversely, changes in the environment influence the agent's actions (reactivity). Part of this environment is made up of other agents. Agents can act to influence each other by sending messages (social behaviour), or do so indirectly via the effects of other actions.

Implicit in all of this is the potential for agent adaptation. For instance, if an agent is to operate with true autonomy in a complex environment, it has to react to a spectrum of circumstances that cannot be foreseen individually by the agent's designer. Autonomous agents may need to explore alternative reactive and proactive strategies, evaluate their performance on-line, and formulate new, innovative strategies without user intervention.

A considerable amount of research on multi-agent systems has been devoted to means of co-ordinating the actions agents, for example via negotiation protocols such as contract net. Such co-ordination takes place over a relatively short timescale, typically in the set-up and enactment of one process instance. We are interested in phenomena operating over much longer time-scales arising from the cumulative effects of repeated interaction of this

sort, i.e. in the potential for simple adaptation at the individual level to engender complex adaptive behaviours as an emergent effect at the system level. The result of each agent exerting, responding to and adapting to local forces, is a dynamic, malleable organisational structure that itself adapts to environmental forces at a higher level.

This begs the question:

- How does one obtain desirable, emergent, adaptive behaviour at a system level, in a world where large numbers of distributed, self-interested agents are interacting in complex, massively parallel, and only partially regulated ways?

Or, more specifically:

- What mathematical and empirical tools do we use to understand the emergent computation in such systems?
- How do we use the information obtained from these tools to design, alter, adapt and control the behaviour of individual agents in such systems, such that desirable global, emergent behaviour of the system is obtained?
- What new applications can be designed around such systems and techniques?

Although this paper clearly cannot answer these questions, it is hoped that the work presented here suggests some new research directions for providing answers.

The emergence of complex adaptive behaviour and self-organisation has been studied in a variety of disparate fields, with a surprising degree of universality observed in the emergent phenomena (see for example the papers collected in Cvitanovic [8], and in Forrest [9]). These fields include evolutionary biology and the 'Santa Fe school' of economics (see, for example, Arthur et al [10]). The first of these has given rise to the paradigm of evolutionary computation (EC [11]). The analogue in the second case is less well defined, though there is a growing community of multi-agent researchers investigating emergent effects in systems of 'economic agents' [12].

At first, evolutionary and economic paradigms may seem quite different. However, one can define a market as a system of self-seeking, profit-maximising agents (traders) exchanging goods and/or services. One can define an evolutionary system as one where a population of self-seeking agents exchange genetic material, and strive for survival, in order to reproduce. In this light, the similarities are obvious. Kearney and Merlat [4] directly address the issue of market dynamics in agent-based systems. This paper presents preliminary work examining the interplay between the two techniques, but firstly the motivations for

the EC paradigm in relationship to agents is looked at more closely.

3. Motivations for the EC models in agent-based systems

Areas where agents could benefit from adaptation are addressed by active research in machine learning (e.g. classification of unforeseen inputs, strategy acquisition through reinforcement learning, etc). However, many machine-learning techniques are focused on centralised processing of databases to formulate models or strategies. In contrast, EC techniques are inherently based on a distributed paradigm (natural evolution), making them particularly well suited for adaptation in systems of agents.

Consider the first author's previous work on acquiring novel combat manoeuvre strategies for fighter planes [13] using genetics-based machine learning (GBML). In this project, systems of distinct rules co-evolve under the action of an EC system to specify complex manoeuvres. This work shows that complex, innovative, multi-component adaptive systems can emerge from EC processes. Note that this is a system-level emergent behaviour, created by the self-interested evolutionary behaviour of many individual rules (agents).

Empirical observations from previous work are compelling. However, it is useful to examine how EC is theoretically motivated as an approach to agent-based systems.

3.1 EC theory's relationship to agents

One of the foundations of EC theory is John Holland's seminal book [14]. This book is often cited as the first on genetic algorithms (GAs, one form of EC) [15], although it never directly mentions GAs. The book is actually directed at general issues of adaptation in systems of distributed, self-interested entities. Holland mathematically examines how populations of self-interested, autonomous entities interacting in systems can, and should adapt, for system-level, desirable, emergent effect.

Agent interactions as a genetic adaptive (EC) plan

To understand how Holland's theories relate to agent-based systems, assume:

- an encoded feature set can describe the make-up of an agent,
- each agent interacts with others through some sort of messaging — further, assume that an agent gathers information about the features of other agents via these messages.

Note that there is seldom a clear mapping from an agent's features to the agent's utility. Therefore, how one approaches the propagation of features through a population of adapting agents is unclear. Holland's theories address this problem.

If agents use the features that are passed via messages as a basis for changing their structure, or the structure of new ('children') agents, this is a description of a form of Holland's genetic adaptive plan, which is a fundamental form of EC. In broad terms, Holland's genetic adaptive plans consist of the following characteristics:

- a set (population) of structures (agents) upon which to operate,
- selection, which is any set of operations under which higher utility agents have higher probabilities of surviving and reproducing,
- a set of 'genetic operators' — these can be of a general character, but usually include some form of:
 - recombination, which can be described as any set of operations where one agent's features are combined with those of other agents,
 - mutation, which can be described as any set of operations where, with some small random probability, agent features are changed at random.

Any plan for altering the structure of a population of agents that includes these characteristics qualifies as a genetic adaptive plan. The genetic analogy of 'survival' and 'reproduction' need not be taken literally. Schemes that change agent structure can be looked upon as causing agent's 'death' and 'rebirth' in some other form, thus making the scheme fall under the rubric of genetic adaptive plans.

In Holland's discussion of genetic adaptive plans, the 'agents' are (natural or artificial) genetic actors, the encoded features are genes, and messages are analogous to gametes. It is important to note that although Holland does not explicitly use the term 'agent' in this sense, his theories deal directly with systems of distinct, adaptive, self-interested entities. Therefore, the basics of Holland's adaptive system theories can be discussed directly in terms of agents, as is done in the following section.

The theory of genetic adaptive plans

There are four main theoretical pillars upon which Holland's genetic adaptive plan models are based. Although these pillars are not discussed in mathematical detail here (for the sake of brevity), their implications are discussed in an agent-based context below.

The **k-armed bandit argument** shows that it is optimal for the number of agents containing an apparently desirable feature to increase exponentially, with respect to the time used to evaluate that feature. An exponential rate of propagation (as opposed to linear, geometric, or super exponential rates) achieves the best compromise between the risk of premature adoption on the basis of misleading evidence, and being overly cautious. The result applies regardless of the distribution of any noise surrounding the feature's apparent utility.

The **schema theorem of GAs**, shows that genetic adaptive plans result in just such an exponential propagation of features. A simple statement of this theorem is of the following form. Consider any given agent feature to be a schema, called H . Then:

(Expected number of agents containing H at time $t+1$) $\geq$

(Number of agents containing feature H at time t) $\times$

(Expected rate of increase (due to selection) of the number of agents containing H) $\times$

(Probability of an agent losing H , due to recombination and mutation)

Note that the process implied will lead to an exponential increase in the number of agents containing any given feature H , if agents containing H are consistently favoured by selection, and if H is likely to survive recombination and mutation. Features that meet this exponential-growth criteria are often called building blocks.

Also note that no features are ever explicitly evaluated under genetic adaptive plans. Instead, selection, recombination, and mutation act at the agent level only.

In other words, distributed, genetics-like interactions between agents (mating, exchange of feature information, recombination of features, and low-probability random mutation of features) lead to an exponential rate of propagation of apparently desirable features through a population of agents. Thus, genetic adaptive plans provide a (provably) near-optimal way of propagating features through a population of agents.

The **building block hypothesis** is a heuristic that assumes the following. If one can propagate advantageous feature (building blocks) in a population of agents, recombination effects will combine these features to build improved agents. Although this heuristic is clearly not correct in all situations, it is bolstered by the following.

The **implicit parallelism** argument shows that, in applying genetic adaptive plans to a population of agents, one implicitly applies the k -armed bandit argument, through

the schema theorem, to a large number of building blocks, to which the building block hypothesis might accurately apply. Clearly, one cannot expose each and every feature to the same exponential decision process implied in the schema theorem. However, the implicit parallelism argument shows that for moderate population sizes N , the number of building blocks (agent feature sets) being processed is on the order of N^3 , i.e. genetic adaptive plans implicitly exploit the parallelism of a distributed, agent-based system to evaluate and propagate a large number of features in a near optimal way.

Given these four arguments (as well as more modern GA theories, especially those concerning co-evolution, niching, and speciation), EC plans seem the appropriate way to propagate features through a system of agents, and to analyse the propagation of agent features in a general context.

3.2 Software embodiment

Although the type of adaptive plan suggested above is theoretically defensible as being appropriate for general agent interactions, it is important to note that most extant EC systems are not decentralised, or agent based. One must consider the implications of this for the use of the EC paradigm in general agent systems.

To implement an EC system, a programmer would typically write computer code that creates a population of EC software entities, and manipulates those entities in a pre-specified way. This code, and its pre-specifications of behaviour, are a sort of 'bubble' around the software entities. In other words, typical EC systems are centralised. Although EC theory typically maintains a focus on decentralisation, typical systems achieve EC effects via a centralised, god-like program. Any emergent behaviour takes place within the constraints of this centralisation.

Even in decentralised EC systems, like Tierra [16], there are server programs that control and pre-specify the nature of entity interactions, in a way that is entirely specific to the simulation. This complicates the relationship between typical EC and real-world decentralised applications. In other words, it is possible one cannot examine the effects of EC in such systems, without considering the real-world computational complications related to such systems.

This observation is related to the argument for embodiment in robotics [17]. This argument would state that one cannot remove the intelligence of an entity from the sensors, actuators, and environment of that entity. The philosophy of embodiment has led to the novel, successful, intelligent control systems that are closely coupled to the noise and complexity of real sensors, actuators, and environments [17].

In the software world, there is a similar argument for situated intelligence. The agent-based EC system presented in the remainder of this paper adopts a similar philosophy of software embodiment. This manifests itself in two ways:

- the system presented is designed to be composed of distinct, autonomous agents, and their behaviours, rather than around controlling systems or servers,
- the environment in which the agents exist is modelled on standard software infrastructure (including agent-based systems standards) — although any implementation cannot consider the effects of every possible agent-based system, it is hoped that adherence to some standards will provide more general conclusions than could be obtained in centralised agent system simulations.

4. Embodied EC agents

4.1 Early results

Given the previous motivations for considering EC as a paradigm for adaptation in agents, it is useful to examine basic results with a system of agents that exchange genetic information to adapt their behaviour.

Previous work introduced an agent-based EC framework [18]. Details of the system are not included here,

for the sake of brevity, but can be provided through contact with the authors.

In Smith and Taylor [18], the framework was tested on simple optimisation problems. In these tasks, the utility of one agent does not depend on the utility of any other agents. Consequently, these tasks do not require co-evolution. Results from one such experiment are shown in Fig 1.

In this experiment, each agent was set with the simple task of finding fit mates. Every agent advertises (through a Plumage object) its own fitness as the number of ones in an internal, 32-bit chromosome. Every agent trusts the fitness advertised by other agents. Note that such trust is not necessary for the general operation of the framework.

The result in Fig 1 shows that, despite the asynchronous, decentralised, agent-based nature of the system, the behaviour is like that of a standard EC system. The distributed nature of the experiment is better illustrated in Fig 2.

In this experiment, when an agent creates another ‘child agent’, it ‘dies’. Thus, the progression shown in Fig 2 can be thought of as a single, immortal, evolving agent.

Figure 2 shows the fitness of a single ‘maternal line’. Imagine that this line of agents remains the ‘property’ of a

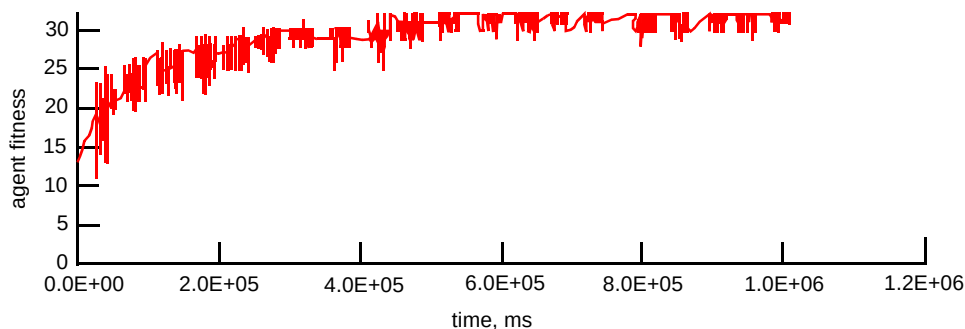


Fig 1 A result with the EC agent framework applied to a simple optimisation task.

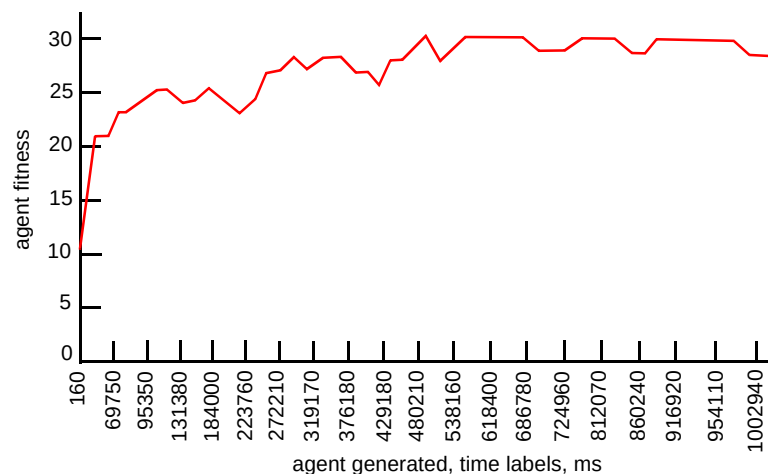


Fig 2 A single maternal line, from the experiment shown in Fig 1.

single user. This user's agents could have genetic interactions with those of other users, through the framework. Such interactions would improve those agents belonging to the user.

This and other experiments show that, as expected, the EC agents behave in a similar fashion to centralised GAs when applied to typical GA problems. The research presented in the following sections begins a deeper investigation, where evolving agents are interdependent. There are proven theories on how an EC system can implicitly balance a diverse, co-operative population, while exploiting Holland's theories [19—21]. However, these issues have yet to be evaluated in a standard framework of asynchronous software agents. The work presented below sets the stage for this examination, in a producer/consumer agent world.

4.2 The producer/consumer interaction framework

Much work has been done in the area of co-evolution in EC systems [22]. However, many of the systems in past research are difficult to parameterise for detailed examination of various effects. This paper presents a much more tractable system for these examinations, based on producer/consumer economics. Note that since many problems of decentralised management and control can be framed as markets, this is far from being an artificial setting.

The problem setting is a variant on that described in Kearney and Merlat [4]. Any number of agents interact in a specific economy. Each agent in the system has a pre-specified number of workers at its disposal. These workers can be allocated to any one of M possible technologies. There are N possible goods in the economy. Each technology converts one set of goods into another set. Technologies are a pre-specified aspect on the economic world.

Note that many decentralised management problems can be cast in this form. A technology is simply a way to perform some task with certain characteristics — input requirements to achieve a given level of output, unit cost as a function of production rate, etc. Workers are simply a means of varying the production rate subject to a constraint due to the finite availability of a resource. The goods in question could be (typed) work items in a workflow management application or packets in a data network.

From an agent's perspective, the goal is to allocate its workers, and maximise its profit. An agent must perform this task, given that the price of each good varies with time, with the action of other agents, and with external market forces. Each of these price variations takes place through a market process of supply and demand. From a scientific perspective, one goal is to demonstrate the evolutionary emergence of interesting, productive economic interaction

between selfish, co-evolved agents. Another goal is to examine variations in emergent behaviour, based on variations in system parameters, and individual agent behaviour. Note that Kearney and Merlat [4] address a similar goal using market mechanisms solely. Our aim is to combine the two strands of work in the future.

Understanding how interaction mechanisms (genetic, economic or otherwise) can be used to cause a collection of autonomous entities to behave as an intelligent, adaptive system is the key to resolving the local autonomy versus central control paradox highlighted earlier. Using such mechanisms in its management and software infrastructure will enable BT to retain the benefits of being a large organisation while becoming agile enough to compete with small competitors and new entrants into its various markets.

4.3 Software structure

The producer/consumer system presented here is intended as a platform for a wide variety of ongoing experiments. Therefore, it was important to maintain an extensible, object-oriented design. The following are the key classes in that design. Like the agent-based EC framework, the agent classes here are built around IBM's Aglets [23]. However, they could be easily transferred to other agent systems. Each class is extensible for future experimentation.

The marketAgent class

Agents of this class process orders for goods from producer/consumer agents, and credit these agent's cash account.

The economicWorld class

This class defines the 'rules' of the economic world, including:

- M , the number of technologies,
- N , the number of goods,
- the set of technologies, what they do, and,
- external consumers and producers (of end goods and raw materials, respectively).

The producerAgent class

This is the key agent class. Each producerAgent responds to 'current prices' messages from agent(s) of the marketAgent class. The producerAgent determines a worker allocation based on this current information, and its own Genotype object (see details of the EC agent framework in Smith and Taylor [18]). The producerAgent then submits an order message to the marketAgent, with negative and

positive quantities representing sell and buy orders for each good.

When triggered by an internally defined condition, a *producerAgent* searches out mates, and attempts to produce child *producerAgents*.

Each of these basic classes is extensible for a number of market and genetic behaviours. Also, there are sets of utility classes that allow for graphical interaction with an experiment

5. Preliminary experiments

To illustrate the basic operation of the *producer/consumer* economic framework, a simplified economic world was constructed.

5.1 The market

In this world, there is only one *marketAgent*, which maintains a store of each good. Each store can have positive and negative quantities, and represents the excess supply of a good. The effect of supply and demand is simulated, by basing each good's price, P , on a simple function of the amount in the *marketAgent*'s excess store of that good, S :

$$P = \exp(-\lambda S)$$

where λ is a parameter.

Prices are only recalculated at the end of a trading day. A trading day begins with the *marketAgent* broadcasting a current price message to every *producerAgent*. The market then asynchronously processes orders from each *producerAgent* that received this current price message. The trading day is complete when all these orders are processed. Prices are then recalculated, and a new trading day begins.

5.2 The agents

In this preliminary experiment, the *producerAgents* are also simplified. In its *Genotype*, each *producerAgent* has a *BooleanGene* for each of the M technologies. A *producerAgent* allocates its workers evenly to each technology that is marked true in its *Genotype*.

In the preliminary experiment, *producerAgents* trade for 10 trading days, and then they (asynchronously) broadcast and receive *Plumage* and *Sperm* objects, to locate mates. When five or more *Plumage* objects are received, the best is picked, and the associated *Sperm* object is used to create a *producerAgent* child. Then the maternal parent agent dies. Since the parent is replaced one-for-one by its offspring, one can consider the maternal line as a single, immortal, adaptive agent.

5.3 The economic world

In the preliminary experiment, it was desirable to have a set of technologies and goods that interacted in a uniform, extensible manner. Therefore, the following scheme was employed. Goods were 'stacked' into R rows and C columns, as shown in Fig 3.

In Fig 3, the unfilled arrows represent technologies. Note that each technology takes two goods in one row, and converts them into a single good in the next row. Such conversions are conservative, such that x quantity of one good, combined with y quantity of another good, yields $(x+y)$ quantity of the third good. Note that the world 'wraps around' at the right and left edges forming a cylinder. Thus, the world is expandable via the R and C parameters, and goods in the $R-2$ intermediate rows are treated identically. The cylindrical topology also avoids edge effects, and means that the environment of each agent is more similar to that experienced in a larger community.

The filled arrows in Fig 3 represent external suppliers of raw materials, and consumers of end goods. The raw material goods (in the lowest row) and the end goods (in the highest row) are controllable through the supply and demand of external suppliers and consumers. Note that the total number of goods $N=R*C$, and the total number of technologies $M=(R-1)*C$. The preliminary experiment presented here is the minimal meaningful form of this world. There is one layer of intermediate goods ($R=3$). For simplicity, $C=3$, giving a total of $N=9$ goods and $M=6$ technologies. External supplies and demands are manipulated such that there is a fixed price for the raw materials, and another fixed price for end goods.

5.4 Results

In the first experiment, the prices of these raw materials and end goods are both maintained at a value of one. A population of 50 *producerAgents* interact in the market, and in the EC mating process.

Prices of the three intermediate goods in this simulation are shown in Fig 4. Variations in these prices are due to evolution of the agents involved in the market. As prices for intermediate goods rise (due to low supply and high demand), agents evolve to produce these high-profit goods. This causes supply to increase, demand to fall, and prices to fall. Then the agents evolve towards being consumers of the low-cost intermediate goods. Note that the agents involved have no memory of price history.

In a second experiment, the price of raw materials remains fixed at 1, but the price of end goods is raised to the fixed value of 5. Thus, a fixed price gradient is maintained from raw materials to end goods. Prices of the three intermediate goods in this simulation are shown in Fig 5.

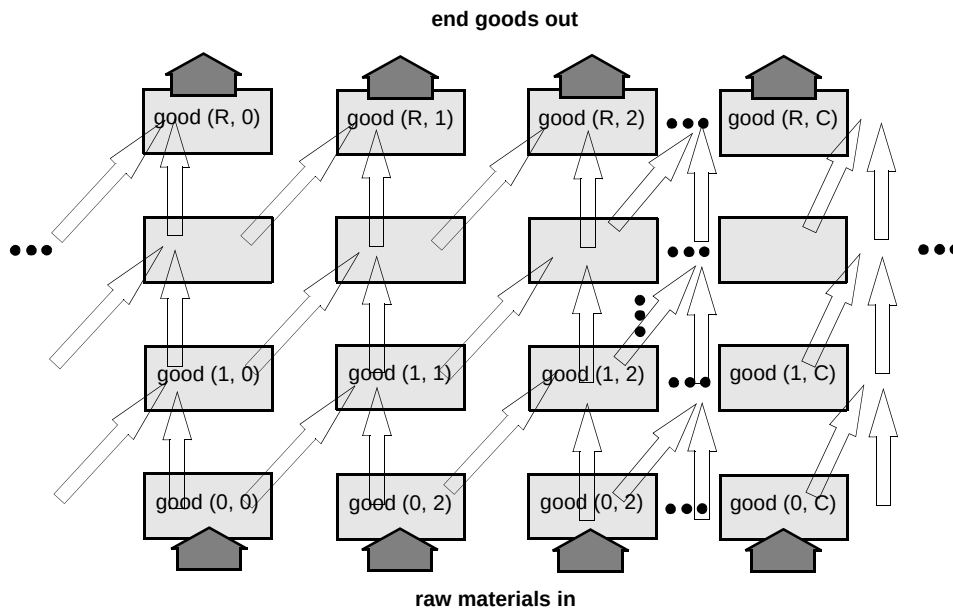


Fig 3 The simplified economic world used in preliminary experiments.

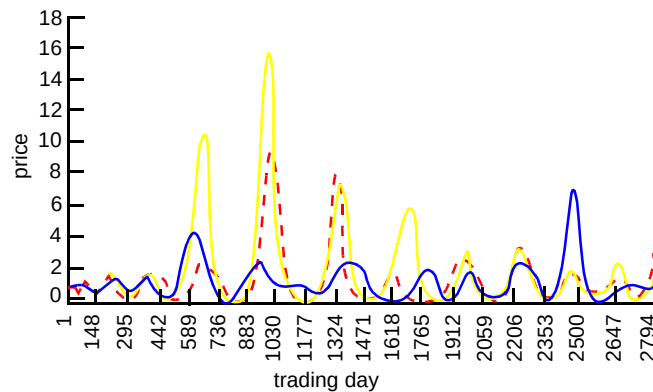


Fig 4 Prices of three intermediate goods as a function of a trading day in an evolving economy of producers and consumers.

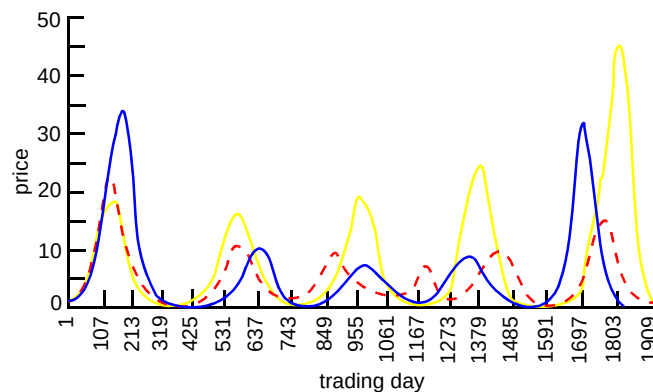


Fig 5 Prices of three immediate goods as a function of a trading day in an evolving economy of producers and consumers, with a fixed, positive price gradient

Clearly, these are only preliminary results from the system. However, they do point to the advantages of the framework presented here. Moreover, this system yields more tractable results than those used in extant research on co-evolving systems (for instance, Smith and Dike [13]). Consider, for example, a Fast Fourier Transform of the data

in Fig 4, shown as a power spectral density in Fig 6, where the power spectrum for each good is normalised, such that the maximum power equals one.

The graph shows the complexity of the price fluctuations for each good (i.e. there are many significant

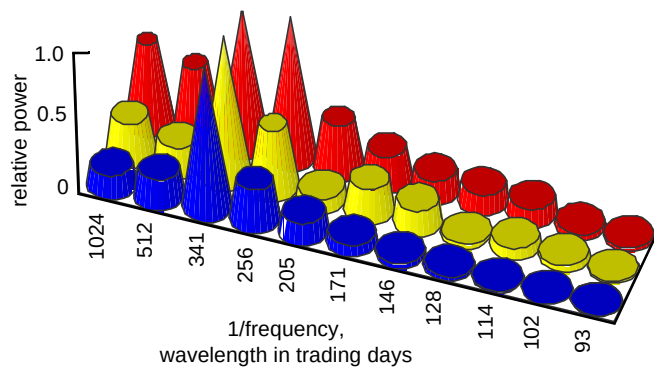


Fig 6 A power spectral density of prices for the three intermediate goods shown in Fig 4. Power values are normalised, such that the maximum power for each good equals one.

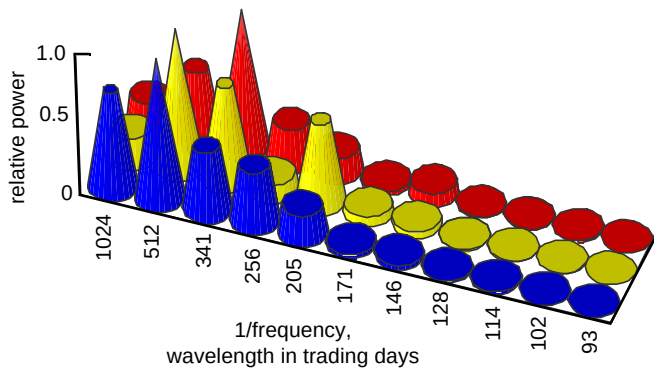


Fig 7 A power spectral density of prices for the three immediate goods shown in Fig 5. Power values are normalised, such that the maximum power for each good equals one.

frequency components). Note that, given the symmetries of the economic world, there is no clear reason that each good should have different price variations. However, although the goods all share some frequencies of variation in common (chiefly those around 1/300 trading days), there are broad variations in the secondary frequency components. Contrast these results to those shown in Fig 7, which is a similar power spectral density for the experiment shown in Fig 5. Recall that in this experiment, there is a positive price gradient maintained from the raw materials to the end goods. The power spectral density shows that the price variations in this experiment are much less complex than those in the previous experiment (i.e. there are fewer pronounced secondary frequencies). Moreover, the price variations in the intermediate goods (although different in scale) are more similar than those of the previous experiment (i.e. there are fewer differences between the spectra of the different goods).

Although these results are far too preliminary to imply general conclusions, they may indicate that the random 'genetic drift' that can occur in EC systems may be encouraged by a lack of opportunities for profit, causing more unstructured behaviour, as in Fig 6, as compared to Fig 7.

6. Future study

Clearly, these experiments only begin to scratch the surface of a large area for investigation that is of importance to both the EC and the agent-based systems communities. The research environment presented here is uniquely suited to this investigation. It has the advantages of:

- embodiment of the agents in a generalised, fully asynchronous agent system (IBM's Aglets),
- a flexible, object-oriented design, well-suited to modifications, extensions, and further experimentation,
- possible extension into real-world applications in the longer term, such as decentralised workflow management systems, IP networks employing usage based and differential charging, supply chain and enterprise resource management,
- co-evolved results that manifest themselves in a far more tractable form than in previous genetics-based machine-learning research.

The last point is key — previous research efforts on co-evolved systems have used experimental environments where clear-cut examination of agent interactions was difficult or impossible.

Several further experiments with the current system present themselves immediately, including those discussed below.

6.1 Problem variations

In the preliminary experiments shown here, supply and demand are manipulated to maintain constant prices for raw materials and end goods. However, a more realistic simulation would consider fixed flow rates for raw materials and end goods, or flows that are dependent on prices. This is a fertile area for exploration, which is easily supported by the current framework.

Another problem variation lies in the complexity of technology interactions. In the preliminary experiments presented here, there is a uniformity of the economic world, induced by the similarity of all technologies. Breaking symmetries in the economic world, through the introduction of more complex technologies, is an interesting area for investigation. Once again, this extension is easily introduced into the current framework.

6.2 Agent variations

The most significant simplification in the preliminary experiments is that the agents assign their workers blindly, as dictated by their genetic code. All reactivity to price patterns occurs on the evolutionary time-scale. Contrast this with the work presented in Kearney and Merlat [4] which has no genetic component and in which adaptation and co-evolution occurs as a result of reactive response to the market environment.

One next step in our research is to combine the two bodies of work. As a first step, this would involve no explicit learning of market patterns during the agent's lifetime. Note, however, that in adjusting internal parameters (levels of bids, etc) in response to market conditions that themselves are influenced by its own past bids, an agent is undergoing a form of reinforcement learning. The agent would only have the capacity to shift worker allocation in reaction to current conditions, due to a strategy dictated by its genetic code. This closely parallels previous work on co-evolution in learning classifier systems [1].

A second step would involve explicit memory of past price patterns within the agent, and resulting lifetime learning. Through such learning, an individual agent would have the capacity to predict and anticipate price patterns. Once again, this closely parallels previous work [1]. An initial investigation using the EC mechanisms to select between a range of predictive and reactive strategies in a similar market context has already been performed by the Dynamo team. Note that as the complexity of the agent's strategies increases, issues of embodiment become more

important. An agent's ability to sense, calculate, and execute are implicitly tied to its computational interactions with other agents, and interactions with its computational environment. This can be seen in the Kearney and Merlat paper [4], where the rules of behaviour, the bid structure and market mechanism all interact to produce the agent and system behaviours. In these situations, the sort of framework presented here is key to future investigation. The current framework will easily support these extensions.

6.3 EC variations

The modifications suggested above lead to important issues in EC. Chief among these is the Baldwin effect. This effect is the interplay of lifetime learning and genetic evolution. If an agent is evolved to learn a particular behavioural strategy, that agent's descendants are more likely to learn that strategy as well, although the strategy is not specifically inherited genetically. This interplay of genetic learning and lifetime learning is key to desirable emergent behaviour in agent-based systems. The current system is well poised for an examination in this area.

Many other more technical EC issues can be examined within the current system, including:

- more complex coding of strategies, including genetic programming techniques [24],
- diploidy and dominance to improve tracking of time variations in the agent-based system [25].

Given the object-based nature of the EC agents framework, these extension are straightforward.

7. Conclusions

The system of producer/consumer agents, coupled with the framework for EC in agent-based systems, presents a clear opportunity for further research. The current system is analysable, while maintaining a realistic, embodied character of the agents. Preliminary results show the promise of investigations using this system. Given that EC seems a theoretically and empirically appropriate technique for developing social, adaptive behaviour in agents, further investigations with this system are promising.

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